

MATH 372: COMBINATORICS THE PROJECT MENU

RELEASED MONDAY, 28 SEPTEMBER 2026 · PROPOSALS DUE FRIDAY, 16 OCTOBER

HOW THE PROJECT WORKS

You will spend the second half of the term on one question of your own, and end with an 8–12 page paper and a fifteen-minute talk at the symposium in finals week. The project is individual. It is the part of the course that is most like a thesis, and it is designed to rehearse the parts of a thesis that are hardest to learn from a problem set: choosing a question that is small enough to finish, running a computation that nobody has told you the answer to, writing up what happened, and talking about it to people who did not do it.

Milestones.

- **Friday, October 16 — proposal and five-minute pitch.** One page: the question, the computation attached to it, what you will read first, and what “done” looks like. The pitch is to the class, in class, and it is the first time you will hear yourself say the question out loud. It incubates over fall break.
- **Friday, November 6 — annotated bibliography and computation prospectus.** Five to ten sources with a sentence each on what you took from them, including the OEIS entries you have found and what they told you; and a description of the computation you will run — what code, what inputs, what you expect to see, and what would surprise you.
- **Week of November 16 — a fifteen-minute meeting with me.** Bring whatever you have. This replaces a written progress report.
- **Wednesday, December 2 — full draft** for peer review. Not extendable, because a late draft spends your reviewer’s time rather than yours.
- **Monday, December 7 — peer reviews** returned.
- **Friday, December 11 — final paper.** Not extendable; the symposium follows.
- **Monday, December 14, 9A.M.–noon, Eliot 207 — the symposium.** Fifteen-minute talks, with discussion, in our exam block.

What the paper must contain. A statement of the question; the mathematics you learned in order to ask it precisely, written for another student in this course; the computation and what it showed; and two components that are graded as content, not as compliance:

- **a section on an approach that failed**, and why — a method you tried that did not work, a conjecture the data killed, a proof that broke at a step you can name; and
- **a computation log**, submitted alongside the paper as an appendix outside the page count: the real record of what you ran, including the dead ends.

A finished paper is always a tidied-up story about the work that produced it; these two components are where the actual work is allowed to show. They are also the two things nobody can produce for you, since there are no false starts if nothing was tried.

An alternative deliverable. One form is allowed in place of the standard paper, at the same mathematical bar: **an interactive artifact** — a web page that lets someone explore the object of

your project, with an accompanying note (four to six pages) containing the mathematics, both left behind on the course page. The course page's visualizations are examples of the genre; yours should let a reader see something a static picture cannot. An artifact is not a lighter option: it still needs a question, a computation, a failed approach, and a log, and it needs to work.

Scope. The test is blunt: *a project that cannot be stated as one question with one computation attached is too big*. Eight to twelve pages is a ceiling as much as a floor, and a small question answered is worth more than a large one surveyed. Every item below has a question and a computation already attached; part of the proposal is to narrow them further to what you can finish by December.

Two exclusions. The project must be new to you. It may not be on the topic of your senior thesis, and it may not be on a topic you are learning or have learned in another course — the Ehrhart theory in item 15, for instance, is off the table for anyone in this term's polytopes course. The point of the project is to find out what you can do with a question nobody has prepared you for, and a topic you are already being taught, or already writing a thesis about, defeats that. If you are not sure whether something is too close, ask; the answer will usually be a way to move the question sideways rather than a no.

How to choose. Read the whole menu once. Pick two or three that you would be willing to think about on a walk. Then look at the prerequisites: some items need material we reach in Acts II and III (marked), which is fine if you are willing to read ahead, and some need only what we have already done. Come talk to me before the proposal; the scoping conversation is the most useful twenty minutes of the project. You may also propose a question not on this menu, if it passes the same test.

THE MENU

The items are grouped by how much is known. *Live* questions are ones where a careful student with a computer has a real chance of seeing something new; the ceiling is high and so is the risk of a negative result, which is a fine outcome if it is written up honestly. *Well-documented* questions have a literature to lean on and a definite thing to learn and compute; they are the straightforward option and there is nothing lesser about them. *Experimental* questions are computations first and theory second.

LIVE

1. Transfer systems on a product of two chains *Acts I–III; the course's own research thread*

A transfer system on a finite lattice is a partial order refining the lattice order and closed under a restriction rule; they arise in equivariant homotopy theory and are counted by Catalan numbers on a chain. On $[1] \times [n]$ the counts 2, 10, 68, 544, ... are known to satisfy a functional equation but no closed form, and on $[m] \times [n]$ almost nothing is known. I have notes and code from this fall summarizing what is known, which I will share with whoever takes this on.

The question. Pick one: (a) how many transfer systems on $[1] \times [n]$ have a given number of arrows, and is that polynomial unimodal for large n ? (b) the horizontal ones are multichains in the Tamari lattice — what are the analogous counts for $[2] \times [n]$? (c) which transfer systems on $[m] \times [n]$ are self-dual under the natural involution?

The computation. Enumerate for small m, n , refine by a statistic, submit the sequence to the OEIS if it is new, and guess a recurrence.

2. Is the poset of unlabeled simplicial complexes Sperner? *Act I only; open as far as I can tell*

The theorem of Chapter 5 says B_n/G is Sperner for any $G \leq S_n$. The poset of simplicial complexes on $[n]$ up to isomorphism is a quotient of a sub-lattice of B_{2^n} , ordered by “is a subcomplex of,”

and the theorem says nothing about it. By computer it is Sperner for $n \leq 5$ (the width equals the largest rank level). Nobody seems to have asked about $n = 6$, where there are 16,353 complexes.

The question. Is $J(B_6)/S_6$ Sperner? Is it rank-unimodal? Does the linear algebra method extend if U is replaced by a weighted operator?

The computation. Compute the width of the $n = 6$ poset (a bipartite matching on 16,353 vertices, feasible), and look for an antichain that beats the largest level.

3. Cyclic sieving *Acts I–III*

The cyclic sieving phenomenon: a set X with a cyclic group action and a polynomial $X(q)$ such that $X(\zeta^d)$ counts the fixed points of the d th power of the generator, for ζ a root of unity. Instances are computer-checkable, the catalogue is incomplete, and the q -analogs of Chapter 6 are exactly the polynomials that show up.

The question. Find a cyclic sieving instance for a family we study — necklaces, order ideals of a product of chains under rotation, noncrossing matchings under rotation — and either prove it or find the smallest counterexample.

The computation. Evaluate candidate q -polynomials at roots of unity and compare with fixed-point counts for $n \leq 12$.

4. Meanders and noncrossing pairings *Act III; connected to current research at Reed*

A meander is a closed curve crossing a line at $2n$ points; the number of meanders grows like $C R^n n^{-\alpha}$ with R and α conjectured but unproven. Meandric systems (several curves) are counted by the entries of the Gram matrix of the Temperley–Lieb algebra’s trace form — the matrix indexed by pairs of noncrossing pairings whose entry is $q^{\text{number of loops}}$ — so the generating function by number of components is the sum of the entries, while the determinant of the same matrix, the meander determinant, has a product formula.

The question. Choose a refinement — meandric systems by number of components, by a symmetry, by a forbidden pattern — and find its generating function or its growth rate numerically.

The computation. Enumerate meandric systems by transfer matrix for $n \leq 14$ and estimate the growth constant by ratio methods.

5. Homotopy types of finite topological spaces *Act I; small and concrete*

A topology on a finite set is the same thing as a preorder, and every finite simplicial complex is weakly equivalent to a finite space. The course page has a map of all spaces on ≤ 6 points with their homotopy types. The minimal finite model of S^n has $2n + 2$ points.

The question. What is the smallest finite space with the weak homotopy type of a given small complex — the torus, the projective plane, $S^1 \vee S^2$? Which types first appear at $n = 7$?

The computation. Enumerate posets on 7 points (2,045 of them), compute cores and homology of order complexes, and tabulate the types.

6. Guess-and-prove *Act III; a skill rather than a fact*

Many sequences in this course satisfy linear recurrences with polynomial coefficients (they are D -finite), and there is software that guesses the recurrence from terms and then proves it (creative telescoping). This project learns the method by using it.

The question. Take a sequence from the course whose recurrence is not in the book, guess it with `ore_algebra` or `gfun`, and prove the guess.

The computation. Generate 50–100 terms of your sequence by a method independent of the recurrence, guess, and verify the guess to more terms than it was fitted on.

7. Cayley's formula five ways *Act I; the classical straightforward project*

The complete graph K_n has n^{n-2} spanning trees. We prove it with the Matrix-Tree Theorem; there are also Prüfer's bijection, Joyal's proof with two marked vertices, the Pitman/Clarke double-counting proofs, and a generating-function proof.

The question. Present three proofs and explain precisely what each one does that the others cannot — for instance, which refinements (trees by degree sequence, forests, trees containing a given edge) each proof gives for free.

The computation. Verify a refinement — e.g. the number of spanning trees of K_n with prescribed degrees $d_1, \dots, d_n$ is the multinomial $\binom{n-2}{d_1-1, \dots, d_n-1}$ — by computer for $n \leq 8$ before proving it.

8. Spanning trees of the cube and its relatives *Act I*

The number of spanning trees of Q_n is $2^{2^n-n-1} \prod_{k=1}^n k^{\binom{n}{k}}$, a consequence of knowing the eigenvalues. The same method handles any graph whose spectrum you know: Cayley graphs of abelian groups, Hamming graphs, Kneser graphs.

The question. For a family of graphs with computable spectra — the two-flip graphs of Problem Set 2 are a natural choice — find the number of spanning trees in closed form.

The computation. Compute spanning-tree counts numerically for the family, factor them, and guess the formula before deriving it from the spectrum.

9. Electrical networks and random walks *Act I*

Resistances, Kirchhoff's laws, and the Laplacian: effective resistance between two vertices is a ratio of spanning-tree counts, and the probability that a random walk started at a hits b before returning is $1/(d_a R_{ab})$. This is the chapter of the text we skipped, and it ties Chapters 3 and 9 together.

The question. Compute effective resistances on a family of graphs (grids, cubes, Platonic graphs) and identify the pattern; or explain Rayleigh monotonicity and use it to bound a hitting time.

The computation. Effective resistance across the cube Q_n between antipodal vertices, for $n \leq 10$, and its limit.

10. Möbius inversion and incidence algebras *Acts I–II; a skipped chapter*

The Möbius function of a poset generalizes inclusion–exclusion and also the number-theoretic Möbius inversion. For B_n it is $(-1)^{|y-x|}$; for divisors it is the classical μ ; for the partition lattice it is $(-1)^{k-1}(k-1)!$.

The question. Compute the Möbius function of a lattice we have met — the partition lattice, the lattice of subspaces of $\mathbb{F}_q^n$, the quotient B_n/G — and use it to count something.

The computation. Implement Möbius inversion for an arbitrary finite poset and check it against the three classical formulas.

11. RSK and the hook length formula *Act II*

The Robinson–Schensted–Knuth correspondence is a bijection between permutations and pairs of standard Young tableaux of the same shape; the hook length formula counts the tableaux. Together they give $n! = \sum_{\lambda} f_{\lambda}^2$.

The question. Prove the hook length formula (the probabilistic proof of Greene–Nijenhuis–Wilf is a good target), or study a statistic RSK preserves — the length of the longest increasing subsequence is the length of the first row.

The computation. Implement RSK, verify $\sum f_\lambda^2 = n!$ for $n \leq 10$, and compute the distribution of the longest increasing subsequence.

12. Chip-firing and the sandpile group *Act I*

Chips on the vertices of a graph, a vertex fires by sending one chip along each edge; the recurrent configurations form an abelian group of order $\kappa(G)$, the number of spanning trees. Reed has a local expert and excellent notes.

The question. Compute the sandpile group of a family of graphs and identify it; or prove the recurrent-configurations-to-spanning-trees bijection for a class of graphs.

The computation. The sandpile group of C_n , K_n , $K_{m,n}$, and the cube, by Smith normal form of the Laplacian.

13. The probabilistic method *any time; classical*

Ramsey's theorem, Erdős's lower bound $R(k, k) > 2^{k/2}$, and the idea that a random object can be shown to exist with a property that no explicit construction achieves.

The question. Present Erdős's bound and one refinement (the Lovász local lemma, or the alteration method), and compare with the best explicit constructions.

The computation. Simulate random colorings of K_n and estimate the probability of a monochromatic K_k for small k , comparing with the first-moment bound.

14. Necklaces, bracelets, and Pólya's theorem *Act III*

Counting orbits of a group on colorings: Burnside's lemma, the cycle index, and Pólya's theorem, applied to necklaces, graphs, and chemical isomers.

The question. Count something the cycle index can count that direct enumeration cannot — unlabeled graphs on $n \leq 10$ vertices by number of edges, or necklaces with prescribed color counts — and compare with the OEIS.

The computation. Compute the cycle index of S_n acting on edges of K_n for $n \leq 8$ and recover the unimodal sequences of Chapter 5.

15. A chapter we skipped, as a starting point *any act; exposition plus a computation of your own*

The text has chapters we will not reach: the BEST theorem on Eulerian tours of digraphs, electrical networks and the cycle and bond spaces, the Ehrhart theory of lattice points in polytopes, and the parts of Young tableaux beyond what Act II covers. Any of them can anchor a project, on one condition: the exposition is the setup, not the deliverable. Re-explaining a chapter of a well-written book is not a project; carrying its main theorem to a computation the book does not do is.

The question. State the chapter's main theorem in your own words, then ask one question it lets you answer that the chapter does not answer — the number of Eulerian tours of a family of digraphs, the Ehrhart polynomial of a family of polytopes, the effective resistance across a family of graphs.

The computation. The computation the question requires, run on enough cases to see the pattern and, where possible, to guess and check a formula.

EXPERIMENTAL

16. The width of the graph poset *Act I*

The theorem says the largest antichain of graphs on m vertices (under subgraph containment) is a middle level. It does not say it is the only one, and for $m = 4$ it is not.

The question. For which m is the middle level the *unique* maximum antichain? What is the size of the largest antichain containing no graph with the middle number of edges?

The computation. Maximum antichains via bipartite matching on the graph poset for $m \leq 7$ (1,044 graphs at $m = 7$).

17. Random walks on quotients *Act I*

The random walk on B_n (flip a random coordinate) descends to a random walk on the quotient B_n/G . Its mixing time and its spectrum are computable from Chapter 3 and Chapter 5 together.

The question. How does the mixing time of the walk on B_n/G compare with that on B_n , for G cyclic, dihedral, and S_n ? Is there a cutoff?

The computation. Simulate and compute exactly the total-variation distance to stationarity for $n \leq 12$.

18. Arrow statistics on Catalan objects *Acts I–III*

The number of transfer systems on a chain with k arrows is the coefficient of q^k in the Carlitz–Riordan q -Catalan polynomial, which is also the area generating function of Dyck paths. There are many other q -Catalan numbers (major index, $dinv$, bounce) and many other Catalan families.

The question. Find the statistic on transfer systems that corresponds to $dinv$, or show that the joint distribution of two natural statistics on transfer systems is symmetric.

The computation. Compute joint distributions of pairs of statistics for $n \leq 9$ and test symmetry and equality with the q, t -Catalan polynomial.

ALTERNATIVE DELIVERABLE

19. An explorer *any act*

A web page in the style of the course visualizations: a Dyck-path-to-transfer-system bijection you can drag, a Laplacian whose spectrum updates as you edit the graph, an RSK insertion you can step through, a sandpile you can topple. Accompanied by a note with the mathematics.

The question. What should a reader be able to *see* that they could not see in a picture? The artifact is judged by that.

The computation. The artifact itself; plus one nontrivial thing you verified with it.

Before you propose. Read the OEIS entries near your topic; email me the two you find most surprising. Then write the one-sentence question and the one-sentence computation, and bring them to office hours or book a Friday slot. The proposal is easier to write after that conversation than before it.